Ulster University

# Human Computer Interaction

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# Introduction

- Human Computer Interface (HCI) was previously known as the man-machine studies or man-machine interaction.
- It deals with the design, execution and assessment of computer systems and related phenomenon that are for human use.
- HCI can be used in all disciplines wherever there is a possibility of computer installation.

# Introduction

- Some of the areas where HCI can be implemented with distinctive importance are mentioned below:
  - Computer Science – For application design and engineering.
  - Psychology – For application of theories and analytical purpose.
  - Sociology – For interaction between technology and organisations.
  - Industrial Design – For interactive products, like mobile phones, *etc.*
- *The world's leading organization in HCI is ACM − SIGCHI, which stands for Association for Computer Machinery − Special Interest Group on Computer–Human Interaction. SIGCHI defines Computer Science to be the core discipline of HCI.*

# Introduction

- The intention of this subject is to learn the ways of designing user-friendly interfaces or interactions.
- Considering which, we will learn the following:
  - Ways to design and assess interactive systems.
  - Ways to reduce design time through cognitive system and task models.
  - Procedures and heuristics for interactive system design.

# Shneiderman's Eight Golden Rules

- Ben Shneiderman, an American computer scientist consolidated some implicit facts about designing and came up with the following eight general guidelines:
  - Strive for consistency.
  - Enable frequent users to use shortcuts.
  - Offer informative feedback.
  - Design dialogs to yield closure.
  - Simple error handling.
  - Permit easy reversal of actions.
  - Support internal locus of control.
  - Reduce short term memory load.

# Strive for Consistency

- Strive for consistency by utilizing familiar icons, colours, menu hierarchy, call-to-actions, and user flows when designing similar situations and sequence of actions.

- Standardising the way information is conveyed ensures users are able to apply knowledge from one click to another; without the need to learn new representations for the same actions.

- Consistency plays an important role by helping users become familiar with the digital landscape of your product so they can achieve their goals more easily.

# Enable Frequent Users to Use Shortcuts

- With increased use comes the demand for quicker methods of completing tasks.
- For example, both Windows and Mac provide users with keyboard shortcuts for copying and pasting, so as the user becomes more experienced, they can navigate and operate the user interface more quickly and effortlessly.

# Offer Informative Feedback

- The user should know where they are at and what is going on at all times.
- For every action there should be appropriate, human-readable feedback within a reasonable amount of time.
- A good example of applying this would be to indicate to the user where they are at in the process when working through a multi-page questionnaire.
- A bad example we often see is when an error message shows an error-code instead of a human-readable and meaningful message.

# Design Dialogue to Yield Closure

- Don’t keep your users guessing.
- Tell them what their action has led them to.
- For example, users would appreciate a “Thank You” message and a proof of purchase receipt when they’ve completed an online purchase.

# Offer Simple Error Handling

- No one likes to be told they're wrong, especially your users.
- Systems should be designed to be as fool-proof as possible, but when unavoidable errors occur, ensure users are provided with simple, intuitive step-by-step instructions to solve the problem as quickly and painlessly as possible.
- For example, flag the text fields where the users forgot to provide input in an online form.

# Permit Easy Reversal of Actions

- Designers should aim to offer users obvious ways to reverse their actions.
- These reversals should be permitted at various points whether it occurs after a single action, a data entry or a whole sequence of actions.
- As Shneiderman states in his book:
  - *This feature relieves anxiety, since the user knows that errors can be undone; it thus encourages exploration of unfamiliar options.*

# Support Internal Locus of Control

- Allow your users to be the initiators of actions.
- Give users the sense that they are in full control of events occurring in the digital space.
- Earn their trust as you design the system to behave as they expect.

# Reduce Short-Term Memory Load

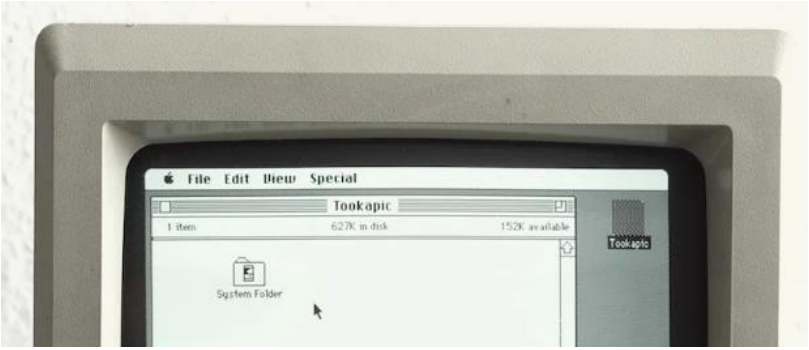
- Human attention is limited and we are only capable of maintaining around five items in our short-term memory at one time.
- Therefore, interfaces should be as simple as possible with proper information hierarchy, and choosing recognition over recall.
- Recognising something is always easier than recall because recognition involves perceiving cues that help us reach into our vast memory and allowing relevant information to surface.

# Examples of Applying Rules

- Apple Inc., a large North American technology company is a great example of how designs reflecting Shneiderman's eight golden rules can lead to successful products.
- The company has achieved great success in everything from their Macintosh to their mobile devices. They pride themselves in their consistent, intuitive and beautiful designs.
- Apple's iOS Human Interface Guidelines, shared publicly in mid-2014, offers a glimpse into how their design team has been applying design principles like Shneiderman's.

# Consistency

- “Consistency” and “Perceived Stability” are woven into the design of Apple’s Mac OS.
- The Mac OS Menu Bar is designed to contain consistent graphic elements regardless of whether it’s a version from the 1980’s or the 2010’s.



# Shortcuts

- As mentioned previously, Mac allows users to use a variety of keyboard shortcuts, commonly used examples include copy and paste (Command-C and Command-V), and taking screenshots (Command-Shift-3).

- Mac allow users to forgo mouse-clicks by providing them with keyboard shortcuts.

# Informative Feedback

- A great example of visual feedback can be seen when a file becomes “highlighted” as the user clicks on a file on a Mac desktop.
- Another example is when the user drags a folder across the desktop, they can see the folder represented as physically being moved as they hold down their mouse.

# Dialogue

- As the user installs software to the Mac OS, an informative screen shows what step the user is currently at in their installation.

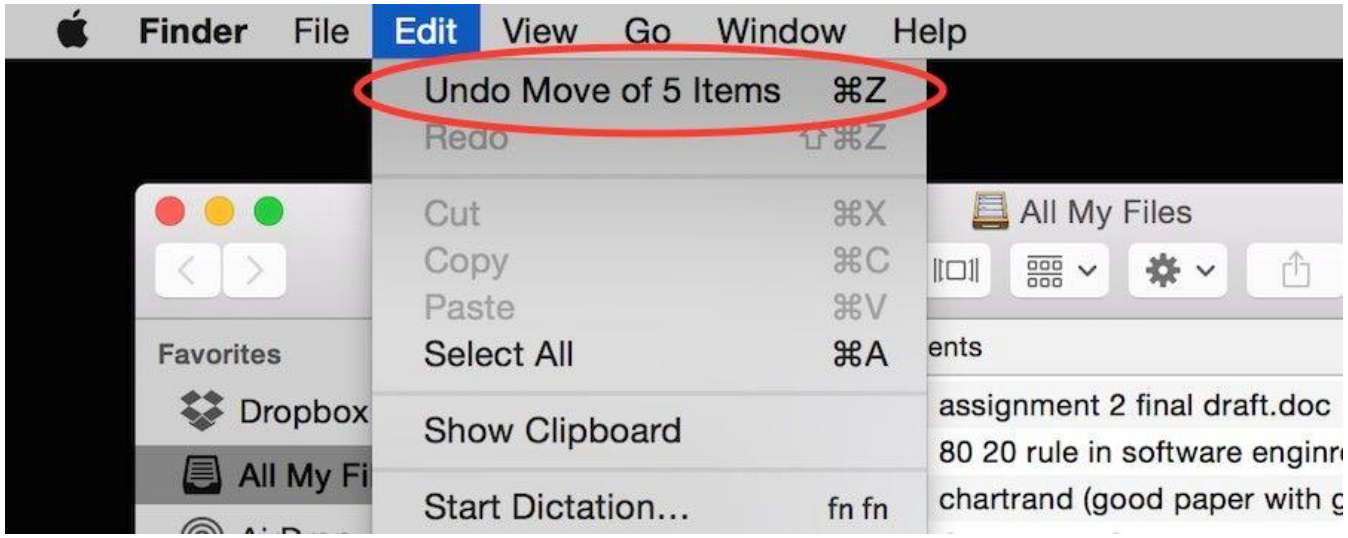
- As the user installs the program “Parallels Desktop 9”, it shows that it is currently “copying files”.

# Error Handling

- During software installation, users are gently alerted with an informative message if an error was made.
- It is important to recognise when to use smaller, less intrusive alerts and when to use greater alerts to warn a user depending on the severity of the error at hand.
- However, it is almost never acceptable to punish the user when errors are made, so be cautious and select the right tone and the right language when drafting an error message that will ultimately be read by your human-users.

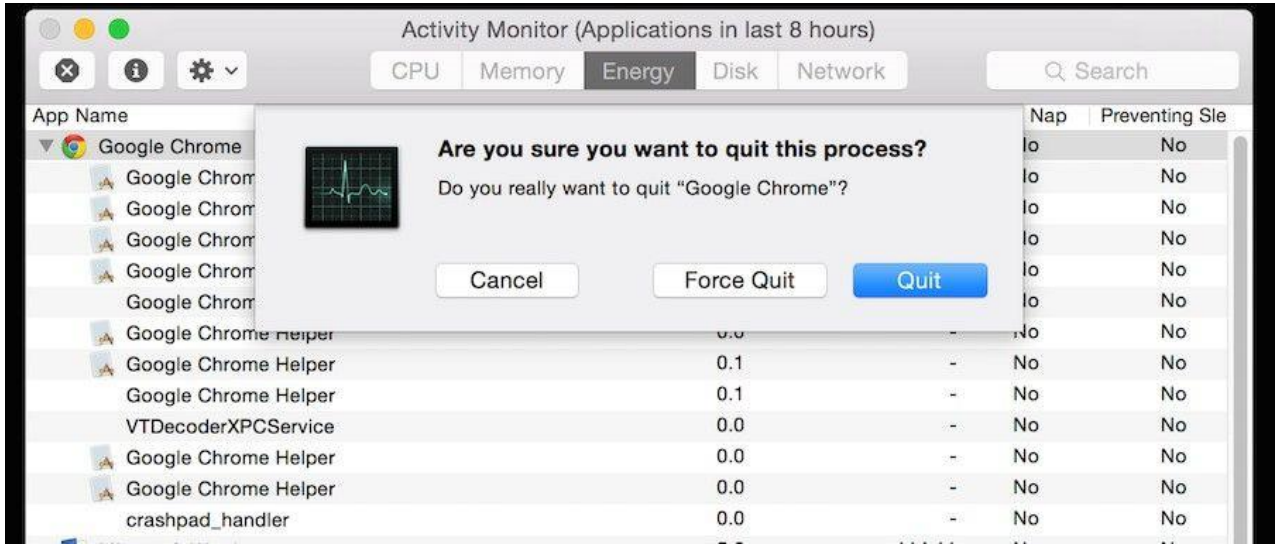
# Permit Reversal of Actions

- When users make an error in providing information during the installation process, they are allowed to go back to the previous step instead of being “punished” by having to start over.



# Support Internal Locus of Control

- Give users the power to choose whether to continue running the program or exit from it.
- Mac's Activity Monitor allows the user to 'Force Quit' when a program has unexpectedly crashed.



# Reduce Short Term Memory Load

- As humans are only capable of retaining 5 items in our short term memory at one time, the Apple iPhone has stuck with allowing only 4 app icons to sit in the main menu area at the bottom of the screen.

- This decision does not only involve consideration of memory load but also considers consistency as well.

# Questions to Ask

## Strive for Consistency

- Is the style of this element maintained across your site/app?
- Is this content placed in the correct location according to the site hierarchy?
- Does this follow the conventions for your chosen platform?
- How can you make your designs more consistent?

# Questions to Ask?

## Enable Frequent Users to Use Shortcuts

- Are there shortcuts available for your more experienced users?
- Who is this product designed for?
- Will there be a need to consider experienced users?
- How can you make it easier and quicker for experienced users?

# Questions to Ask?

## Offer Informative Feedback

- Does the user know where they are at in the process?
- Does the user know what they have done after performing this action?
- How are you communicating this feedback to your user?

# Questions to Ask?

## Design Dialogue to Yield Closure

- Does the user have to do any guessing here?
- Is it clear and obvious enough for your intended audience?
- Are there any next steps for the user?
- How are you communicating the system status with the user?

# Questions to Ask?

## Offer Simple Error Handling

- Have you done everything imaginable to prevent this error from happening on your end?
- Is this error avoidable in the first place?
- If the user does make an error, how easy is it for them to fix it?

# Questions to Ask

## Permit Easy Reversal of Actions

- How many steps does the user have to take to reverse their actions?
- Will the user quickly realize they need to reverse the action in the first place?
- How can you make your users detect the possibility of reversal?

# Questions to Ask

## Support Internal Locus of Control

- Will the user feel in control at this specific touch point in your app?
- Will they be surprised in an unpleasant manner?
- Does the site feel easily navigable?
- Does the user feel safe and in control?
- How can you make the user feel more safe and in control?

# Questions to Ask

## Reduce Short-Term Memory Load

- Are there enough visual cues here for the user to find the functionality or item?
- Do they have to remember things to understand what's going on?
- How can you help the user recall?